

Firdawss El Haddouchi

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About Me

Final-year Data Engineering student at ENSA Al Hoceima with practical experience in AI, Machine Learning, NLP, RAG, and Big Data pipelines. Skilled in building end-to-end data processing systems, semantic search applications, and backend services. Currently seeking a 6-month PFE internship in Artificial Intelligence / Data Engineering.

Experience

AI & Data Engineering Intern – PFA

Al Hoceima, Morocco

Trésorerie Provinciale d'Al Hoceïma – TGR

July 2026 – Sept 2026

- Designed and developed a fully local Retrieval-Augmented Generation (RAG) assistant for administrative document search.
- Engineered end-to-end NLP data pipelines for automated text extraction, cleaning, semantic chunking, and ChromaDB vector indexing.
- Integrated Sentence Transformers for dense vector embeddings and connected a locally hosted LLM with Ollama, FastAPI, and React for document-based question answering.
- **Technologies:** Python, RAG, Ollama, Sentence Transformers, ChromaDB, FastAPI, React

Data & Machine Learning Intern – PFA

Kénitra, Morocco

Centre Marocain de Recherches Polytechniques et d'Innovation (CMRPI)

July 2026 – Aug 2026

- Contributed to the development of an automated threat intelligence pipeline and early detection system for phishing URLs targeting Moroccan SMEs.
- Built a PhishTank data ingestion collector and engineered lexical and domain-based features from raw URL feeds.
- Trained and evaluated Scikit-learn ML models using 5-fold cross-validation, achieving 90%+ precision and recall in detecting malicious links.
- Developed the Statistics and Reflex Sheets sections of the Streamlit prototype and contributed to project integration.
- **Technologies:** Python, Scikit-learn, Pandas, Streamlit, PhishTank, Machine Learning, Feature Engineering

Software Development Intern

Al Hoceima, Morocco

Regional Hospital Center of Al Hoceima

July 2025 – Sept 2025

- Co-developed a web-based radiology department management platform using PHP, JavaScript, and MySQL to digitalize daily operations.
- Built dynamic analytics dashboards using Chart.js to monitor key performance indicators (KPIs) and operational metrics.

Business Intelligence Intern

Al Hoceima, Morocco

Terra Nova EBS

July 2025 – Aug 2025

- Designed multi-page interactive Power BI reporting solutions to monitor business performance across departments.
- Implemented data models, DAX measures, and calculated columns for business analysis.

Projects

TaaSim: Smart Urban Mobility Platform

Mar 2026 – June 2026

[GitHub](#)

- Designed a real-time event-driven Kappa architecture for processing simulated GPS trajectory streams and ride requests.

- Implemented Apache Kafka and Apache Flink stream processing jobs for live mobility metrics and traffic congestion detection.
- Processed historical spatial trajectories using Apache Spark and stored analytics in Cassandra and MinIO.
- Orchestrated and containerized the multi-service distributed pipeline using Docker Compose and Grafana dashboards.
- **Technologies:** Apache Kafka, Apache Flink, Apache Spark, Apache Cassandra, MinIO, Grafana, Docker Compose

JobIntelligent: Big Data Job Analytics Platform

Mar 2026 – May 2026

[GitHub](#)

- Engineered a Medallion Data Lake architecture (Bronze, Silver, Gold) using MinIO and PostgreSQL for job market analytics.
- Automated daily web scraping and ETL ingestion DAGs using Apache Airflow, standardizing data schemas across multiple job portals.
- **Technologies:** Apache Airflow, MinIO, PostgreSQL, FastAPI, Power BI, Python, Docker

Education

National School of Applied Sciences (ENSA)

Al Hoceima, Morocco

Engineering Degree in Data Engineering

2024 – present

National School of Applied Sciences (ENSA)

Al Hoceima, Morocco

Engineering Preparatory Classes in Engineering Sciences

2022 – 2024

Skills

AI & Machine Learning: Machine Learning, NLP, RAG, Sentence Transformers, HuggingFace, ChromaDB, Scikit-learn

Data Engineering & Big Data: Apache Spark, Apache Kafka, Apache Flink, Apache Airflow, Medallion Architecture, Data Lake, ETL/ELT Pipelines

Databases & Storage: SQL, PostgreSQL, MySQL, Apache Cassandra, MinIO, Pandas, Power BI (DAX)

Backend & Development: Python, FastAPI, REST APIs, React, PHP, JavaScript, HTML/CSS

Tools & Deployment: Git, GitHub, Docker, Docker Compose, Streamlit, Grafana

Languages

Arabic & Tarifit: Native

French: Conversational

English: Basic

Courses and Certifications

Supervised Learning with Scikit-Learn: DataCamp

Introduction to Docker: DataCamp